

EventSure – A Full-Stack Digital Event Insurance Platform

The increasing scale and complexity of modern events such as weddings, concerts, corporate gatherings, sports tournaments, and festivals have significantly raised the demand for reliable financial protection against unforeseen risks. Event cancellations, property damage, liability issues, and weather-related disruptions can lead to substantial financial losses. Traditional insurance processes often involve manual policy creation, delayed claim processing, and limited real-time tracking, resulting in operational inefficiencies and reduced transparency.

This project proposes the development of a **Full-Stack Event Insurance Web Application** designed to digitize and streamline the complete insurance workflow. The application will provide a centralized and secure platform where customers, agents, and administrators can efficiently manage event insurance policies and claims.

The system will be developed using a modern technology stack. The backend will be implemented using **ASP.NET Core Web API (C#)**, following a layered architecture to ensure clean code structure and maintainability. Entity Framework Core will be used as the ORM for database operations, with SQL Server serving as the primary data storage system. Secure authentication and authorization will be implemented using JWT-based role-based access control, supporting three primary roles: Admin, Agent, and Customer.

Customers initiate insurance requests for events such as weddings, concerts, corporate gatherings, and sports events. Agents review and process policy creation, while administrators oversee user management and system governance. Once a policy becomes active, it supports payment tracking and structured claim submission. Claims follow a forward-moving lifecycle from submission to review and settlement, ensuring transparency and traceability across operations. The system architecture enforces clean layering through Controllers, Services, and Repositories, maintaining data integrity, secure API endpoints, and organized process flow.

The frontend will be developed using **Angular (latest version)** with TypeScript, providing a responsive and dynamic user interface. The application will open with a dashboard interface, followed by a secure login system that directs users to role-specific functionalities. Angular features such as Reactive Forms, Route Guards, HTTP Interceptors, and modular architecture will be utilized to ensure secure navigation and seamless API communication.

The platform will enable customers to explore event insurance options, purchase policies, and raise claims. Agents will assist in policy creation and customer support, while administrators will oversee user management, policy approvals, and

system-level monitoring. The application aims to improve operational efficiency, enhance transparency in claim processing, and provide a structured digital environment for event insurance services.

By leveraging modern web technologies and secure API-driven architecture, this project demonstrates a scalable, secure, and enterprise-ready solution for event insurance management in a cloud-compatible environment.

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